



ANDERSON JUNIOR COLLEGE
HIGHER 2

CANDIDATE
NAME

--

PDG

--	--	--	--

PDG
INDEX NUMBER

--	--

BIOLOGY

9744/04

Paper 4 Practical

24 August 2017
Thursday

Candidates answer on the Question Paper.

2 hours 30 minutes

READ THESE INSTRUCTIONS FIRST

Write your name and PD group on all the work you hand in.
Give details of the practical shift and laboratory, where appropriate, in the boxes provided.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.
DO NOT WRITE IN ANY BARCODES.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

Shift
Laboratory

For Examiner's Use	
1	
2	
3	
Total	55

Answer **all** the questions.

1	The enzyme lipase catalyses the hydrolysis of triglycerides into fatty acids and glycerol. You are required to investigate the effect on the lipase-catalysed reaction of the independent variables: - enzyme concentration - presence of calcium ions The substrate for lipase will be the triglycerides present in milk. The progress of this hydrolysis can be monitored by using an indicator, T, which changes colour due to the production of fatty acids. You are provided with the following solutions; 25cm³ of milk containing calcium ions, labelled M+C, 25cm³ of milk without calcium ions, labelled M, 25cm³ of indicator solution, labelled T, 30cm³ of sodium carbonate solution, labelled A, 20cm³ of 10% lipase solution, labelled E10, 20cm³ of 5% lipase solution, labelled E5, Lipase is an irritant. You are advised to wear the eye protection provided. Contact of the solution with your skin should be avoided. If it touches your skin, wash it off with tap water. <i>Proceed as follows.</i>	
(a)	Stage 1 Use the beaker or container provided to make a water-bath with warm water, between 38°C and 42°C. Stage 2 Label four boiling tubes, B1, B2, B3 and B4. Using the syringes, put: 2cm³ of solution M+C into each of the boiling tubes labelled B1 and B2, 2cm³ of solution M into each of the boiling tubes labelled B3 and B4, 2cm³ of solution T into each of the boiling tubes labelled B1, B2, B3 and B4 and gently shake, 3cm³ of solution A into each of the boiling tubes labelled B1, B2, B3 and B4 and gently shake so that all the mixture turns blue. Minor variations in colour between the tubes can be ignored as long as the contents are blue. Put the four boiling tubes into the water-bath for at least three minutes, before progressing to Stage 3. Stage 3 After the boiling tubes have been in the water-bath for three minutes, start a stopwatch, which will be left running continuously throughout the investigation. Start and end times will be taken from this stopwatch. Stage 4	[3]

Remove the boiling tubes labelled **B1** and **B3** from the water-bath and put them in a boiling tube rack. Use a syringe to put 2cm³ of solution **E10** into each of these two boiling tubes and mix well. Record the start times in Table 1.1, below.

Stage 5

Observe the boiling tubes **B1** and **B3** and record the times at which the colour changes (end times) in Table 1.1. If the time taken for the colour to change for any boiling tube is longer than five minutes, record 'no change'.

Stage 6

Remove the boiling tubes labelled **B2** and **B4** from the water-bath and put them in a boiling tube rack. Use a syringe to put 2cm³ of solution **E5** into each of these two boiling tubes and mix well. Record the start times in Table 1.1.

Stage 7

Observe the boiling tubes **B2** and **B4** and record the times at which the colour changes (end times) in Table 1.1. If the time taken for the colour to change for any boiling tube is longer than five minutes, record 'no change'.

Stage 8

Calculate the time taken for the colour to change for each of the boiling tubes **B1-B4**, and record this in Table 1.1. If the time taken for the colour to change for any boiling tube is longer than five minutes, record 'no change'.

Table 1.1

	B1	B2	B3	B4
Start time/s				
Time at which colour changes (end time)/s				
Time taken for colour to change/s				

[3]

- Time recorded to nearest second;
- Time recorded to whole numbers;
- B1 and B2 takes shorter time than B3 and B4, reject if B3 records as no change;

(b) Prepare a table in the space below to show the effect of enzyme concentration and presence of calcium ions on the hydrolysis of triglycerides in milk.

Lipase concentration/ %	Presence of calcium ions	Time taken for colour to change/ s	Rate of hydrolysis / s ⁻¹
5	Present		
5	Absent		
10	Present		
10	Absent		

- T1 - Well presented - fully ruled table;
 T2 - Heading with lipase concentration and correct units;
 T3 - Heading with presence of calcium ions, no units;
 T4 - Time taken or rate of hydrolysis recorded with appropriate units;
 T5 - Values brought over and filled in correctly + correct calculation of rate;
 T6 - Consistent number of dp: time and lipase concentration in whole numbers, rate to 3 sf;

[6]

(c) Describe how you could set up a control for the effect of lipase.

Replace lipase with equal volume/ 2 cm³ of distilled water, subject the tube to same experimental

conditions. This shows that a shorter time taken/ no colour change is due to enzymatic action of lipase in breaking down triglycerides.

Also accept: **boiled and cooled** lipase

(d) Identify **one** significant source of error in measuring the dependent variable in this investigation. [1]

Determination of extent of green colour as end point is **subjective** via **visual inspection**.

(e) State **two** ways in which the experimental procedure could be improved.

- Use a colorimeter/ spectrophotometer/ pH probe to determine a **fixed absorbance value/ wavelength of light/ fixed pH value** as end point;
- Use an electrostatically controlled water bath and **monitor temperature using a thermometer**;
- Equilibrate temperature of enzymes E5 and E10 **separately** first before mixing with M and M+C;
- Use a white tile as background for clearer observation of colour change;
- Use a colour chart with end point colour to compare for clearer observation of end point;
- Prepare a negative control boiling tube to compare the colour change against experimental tubes for clearer observation of end point;

Reject:

- Do replicates/ repeats

(f) Explain why the method used in this investigation is not suitable for investigating the effect of pH on the activity of lipase.

- Triglycerides will yield fatty acids and glycerol as products upon hydrolysis/ Fatty acids will reduce the pH of the solution;
- The variable being manipulated (pH) and that being measured (amount of fatty acids to indicate lipase rate of reaction) is the same/ fatty acids will change the pH of the boiling tubes and therefore not possible to keep a constant pH to investigate its effects;

Some students carried out an investigation using lipase and found that its activity was affected by the concentration of copper sulfate solution. All other variables were kept constant.

The results of their investigation are shown in Table 1.2.

Table 1.2

Copper sulfate concentration/ $\times 10^{-3}$ moldm ⁻³	Lipase activity/arbitrary units
1.0	26.0
2.0	12.0
3.0	5.0
4.0	2.5
5.0	1.5

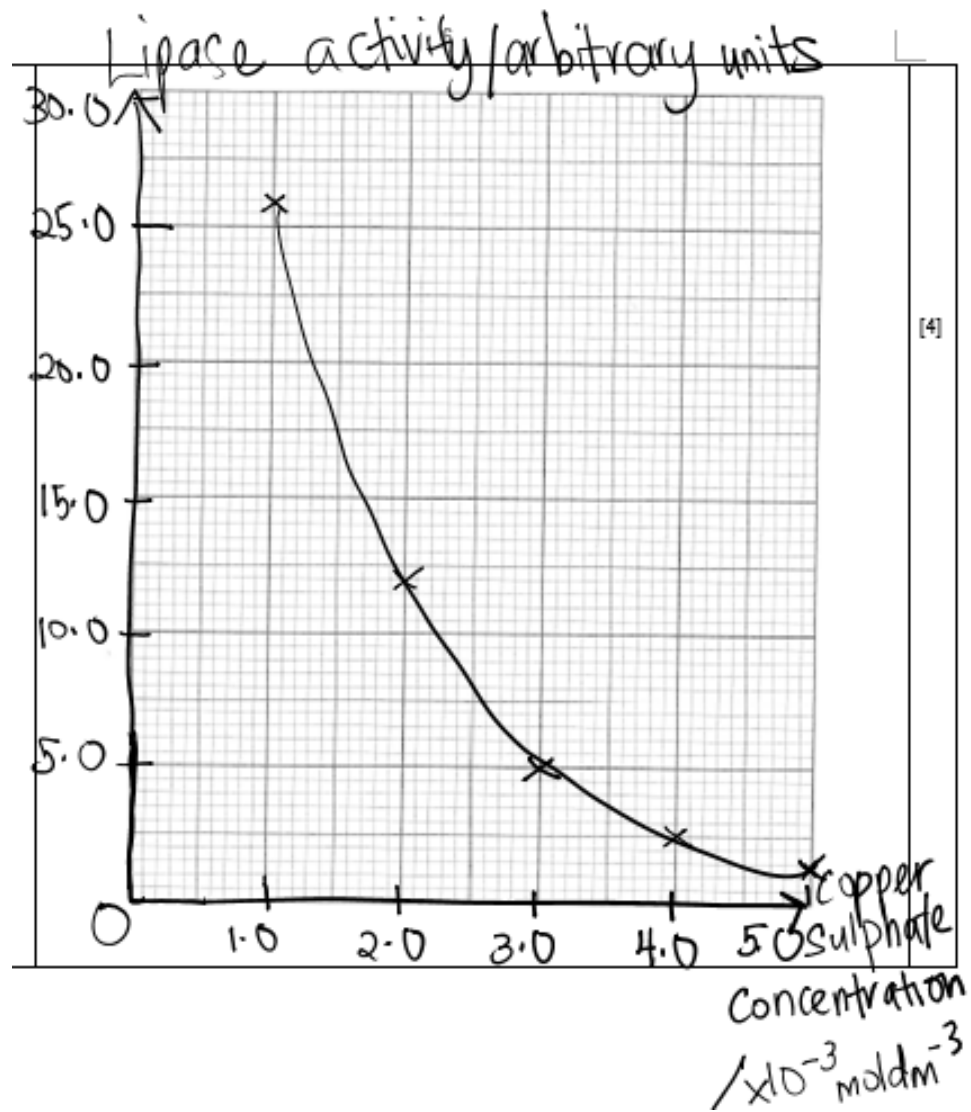
(g) Plot a graph of the data shown in Table 1.2, on the grid below.

- G1 – appropriate scale + graph occupies at least half of graph paper in both x and y axis direction + origin labeled + last y-axis label beyond last plotted point;
 G2 - Correctly labelled X and Y axis + correct units + x and y axis values labeled to one dp;
 G3 - Plot individual points precisely using a cross
 G4 - Line of best fit, Sharp, clear line through/close to the plotted points.

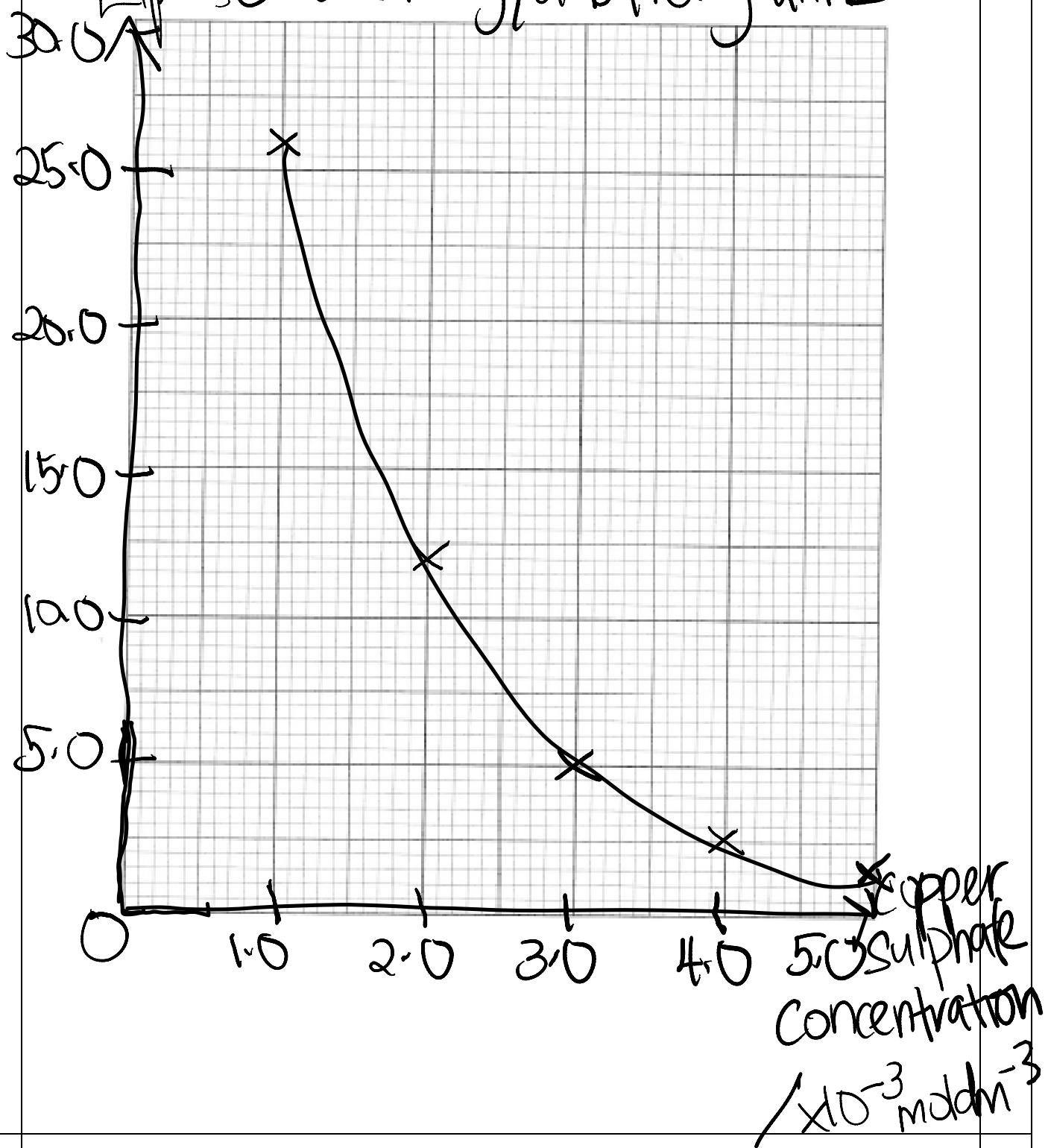
Reject:

- Extrapolated beyond the data
- Unsuitable scale on x and y axis

- Lacked precision in plotting data.



Lipase activity/arbitrary units

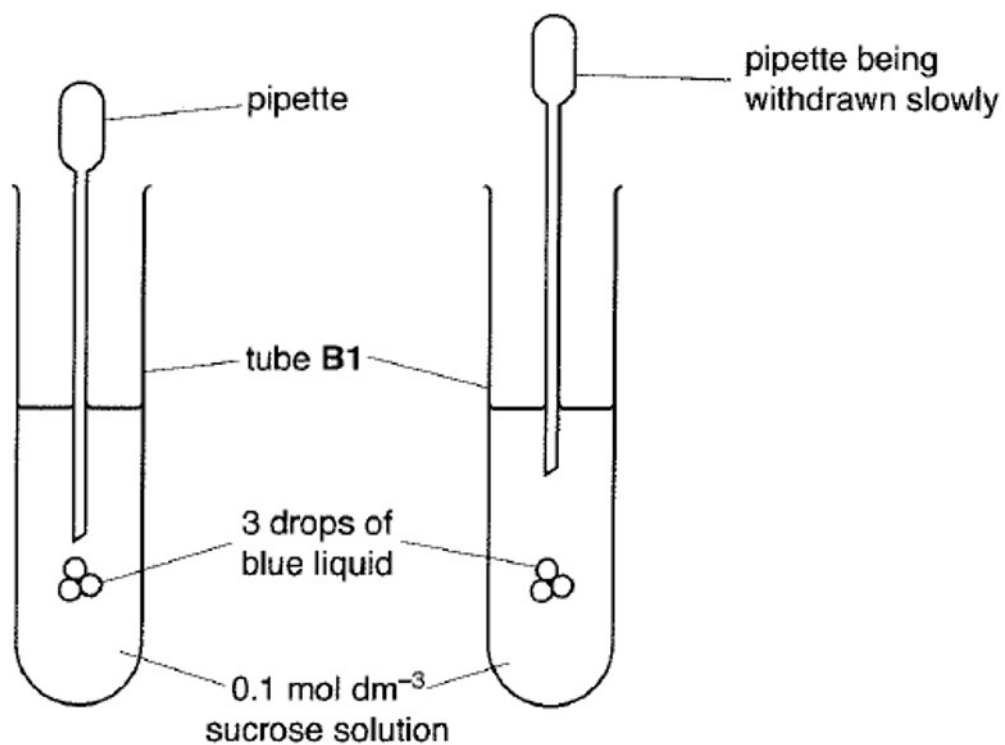


(h) Describe and explain these results.

2	K1 and K2 are stained, transverse sections of leaves from two different species of plant.		
(a)	(i)	Make a large, labelled, plan drawing of K1 to show the distribution of tissues in the leaf lamina (avoiding the midrib). Details of individual cells are not required.	
		UPPER EPIDERMIS PALISADE MESOPHYLL VASCULAR BUNDLE SPONGY MESOPHYLL LOWER EPIDERMIS STOMA	[3]
	(ii)	Make a labelled , high-power drawing to show the detailed structure of three adjacent cells from the palisade mesophyll layer.	
		CELLULOSE CELL WALL PLASMA MEMBRANE NUCLEUS	[3]

	(iii)	Use the stage micrometer to determine the area of the field of view <u>under high power</u> . Calculate the average density of palisade mesophyll cells. State the magnification used and show your working.									
		Density = Average number of cells/ Area of field of view (NOT mass/volume!!) To calculate the area of the field of view <u>under high power</u> (i.e. x400 or x600 only), Use the stage micrometer divisions. At <u>x400</u>, you can see there are 44 or 45 of stage micrometer divisions in diagram below. Each division is 0.01 mm. Hence the <u>diameter</u> of the field of view is <u>0.44 mm</u>. Radius = <u>0.22 mm</u> at x400 Area = $\pi r^2 = \pi (0.22)^2 = 0.152 \text{ mm}^2$ (1 mark) Count the number of palisade mesophyll at least twice. Then get <u>average number of cells</u>. Say you got 112 cells. (1 mark) So density = Average number of cells/ Area of field of view = $112/ 0.152 = 737 \text{ mm}^{-2}$. (1 mark) At x600 magnification, diameter = 30 stage micrometer divisions = $30 \times 0.01 = 0.3 \text{ mm}$ Hence radius = 0.15 mm.	[3]								
	(iii)	Calibrate the eyepiece graticule using the stage micrometer so that you can use it to measure the length along one palisade mesophyll under a suitable magnification. Repeat until you have three measurements. State the magnification used and show your working in calculating the average length.									
		X400 Calibration (1 mark) 40 eyepiece graticule units = 10 divisions on stage micrometer 1 division on stage micrometer = 0.01 mm Hence 40 eyepiece graticule units = $10 \times 0.01 = 0.1 \text{ mm}$ 1 eyepiece graticule = $0.1/40 = 0.0025 \text{ mm} = 2.5 \mu\text{m}$ (1 mark) (1 mm = 1000 μm) Measurement of average palisade mesophyll cell length <table><tr><td>Readings</td><td>Eyepiece graticule units</td></tr><tr><td>1</td><td>19</td></tr><tr><td>2</td><td>18</td></tr><tr><td>Average</td><td>18.5 (1 mark)</td></tr></table> $18.5 \times 2.5 \mu\text{m} = \underline{46.25 \mu\text{m}}$ (1 mark for accuracy) Note: At x 600, Calibration (1 mark) 1 eyepiece graticule = 1.6 μm or 1.6 μm (1 mark) (1 mm = 1000 μm)	Readings	Eyepiece graticule units	1	19	2	18	Average	18.5 (1 mark)	
Readings	Eyepiece graticule units										
1	19										
2	18										
Average	18.5 (1 mark)										
(b)		The plant species from which K2 was taken grows in a dry habitat.									

	Examine K2 , using your microscope.																														
	State four observable features that distinguish K2 from K1 and <u>present the differences in a suitable format</u> .																														
	Any 4 below: <table border="1"><thead><tr><th>Features</th><th>K1 (<i>Ficus</i>)</th><th>K2 (<i>Marram grass</i>)</th></tr></thead><tbody><tr><td>Shape of leaf</td><td>Leaf is flat</td><td>Leaf is curled/ rolled</td></tr><tr><td>Hair-like structures</td><td>Absent</td><td>Present</td></tr><tr><td>Protrusions</td><td>Absent</td><td>Present</td></tr><tr><td>Stomata number</td><td>More</td><td>Fewer</td></tr><tr><td>Location of stomata</td><td>More exposed to air/ lower epidermis</td><td>Not exposed/ sunken/ within grooves</td></tr><tr><td>Air spaces between cells</td><td>More</td><td>Little</td></tr><tr><td>Mesophyll cells</td><td>Elongated (palisade) or round (spongy)</td><td>Round</td></tr><tr><td>Cuticle layer</td><td>2 layers/ on both sides of leaf</td><td>1 layer on one side</td></tr></tbody></table>			Features	K1 (<i>Ficus</i>)	K2 (<i>Marram grass</i>)	Shape of leaf	Leaf is flat	Leaf is curled/ rolled	Hair-like structures	Absent	Present	Protrusions	Absent	Present	Stomata number	More	Fewer	Location of stomata	More exposed to air/ lower epidermis	Not exposed/ sunken/ within grooves	Air spaces between cells	More	Little	Mesophyll cells	Elongated (palisade) or round (spongy)	Round	Cuticle layer	2 layers/ on both sides of leaf	1 layer on one side	[4]
Features	K1 (<i>Ficus</i>)	K2 (<i>Marram grass</i>)																													
Shape of leaf	Leaf is flat	Leaf is curled/ rolled																													
Hair-like structures	Absent	Present																													
Protrusions	Absent	Present																													
Stomata number	More	Fewer																													
Location of stomata	More exposed to air/ lower epidermis	Not exposed/ sunken/ within grooves																													
Air spaces between cells	More	Little																													
Mesophyll cells	Elongated (palisade) or round (spongy)	Round																													
Cuticle layer	2 layers/ on both sides of leaf	1 layer on one side																													
(c)	You are required to investigate some aspects of the water relation of living plant cells.																														
(i)	Label three test tubes B1, B5, B10 and C5 respectively. Using a syringe or pipette, prepare the three concentrations of sucrose using the water and 1.0 mol dm⁻³ sucrose solution provided. Record the volume of water and 1.0 mol dm⁻³ sucrose solution used in the table below. The total volume of different concentrations of sucrose should be 10 cm³. Then place 10 cm³ of sucrose into each of these tubes. <table border="1"><thead><tr><th>Tube</th><th>B1</th><th>B5</th><th>B10</th></tr></thead><tbody><tr><td>Concentration of sucrose solution (10 cm³)</td><td>0.1 mol dm⁻³</td><td>0.5 mol dm⁻³</td><td>1.0 mol dm⁻³</td></tr><tr><td>volume of water</td><td>9.0</td><td>5.0</td><td>0.0</td></tr><tr><td>volume of 1.0 mol dm⁻³ sucrose solution</td><td>1.0</td><td>5.0</td><td>10.0</td></tr></tbody></table>			Tube	B1	B5	B10	Concentration of sucrose solution (10 cm ³)	0.1 mol dm ⁻³	0.5 mol dm ⁻³	1.0 mol dm ⁻³	volume of water	9.0	5.0	0.0	volume of 1.0 mol dm ⁻³ sucrose solution	1.0	5.0	10.0	[2]											
Tube	B1	B5	B10																												
Concentration of sucrose solution (10 cm ³)	0.1 mol dm ⁻³	0.5 mol dm ⁻³	1.0 mol dm ⁻³																												
volume of water	9.0	5.0	0.0																												
volume of 1.0 mol dm ⁻³ sucrose solution	1.0	5.0	10.0																												
	Transfer all 10 cm ³ 0.5 mol dm ⁻³ sucrose solution from B5 into a test tube labelled C5 and add three drops of the dye, methylene blue (labelled MB) to it. Shake tube C5 to make the colour uniform. Suck up a little of this blue 0.5 mol dm ⁻³ sucrose solution into a pipette and then, with the tip of this pipette held stationary, half way down the solution in tube B1 (see diagram below), very gently release three drops from the pipette. Do not squirt these drops into the solution. Withdraw the pipette slowly.																														



Carefully observe the movement of the blue liquid. Repeat the procedure of **B10**.

(i) Record your observations in a suitable format for **B1** and **B10**.

	B1	B10
Observations	Blue liquid disperses and sinks/ remain suspended	Blue liquid disperses and sinks to the bottom/ float.

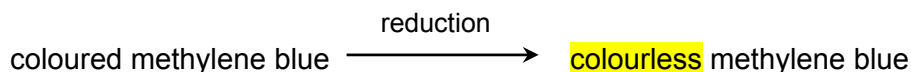
[2]

[Total: 20]

3 Planning question

Yeast undergoes aerobic respiration, breaking down glucose into carbon dioxide and water. This process is catalyzed by enzymes.

Methylene blue is an artificial **hydrogen acceptor** which is blue in the oxidised form and **colourless** when reduced.



A **colorimeter** can be used to measure the absorbance of light of 550 nm by methylene blue solution. When methylene blue is reduced and becomes colourless, its absorbance reading will become **0** arbitrary units. Yeast suspension has an absorbance value of **15** arbitrary units. The colorimeter is shown in **Fig. 4.1** below.

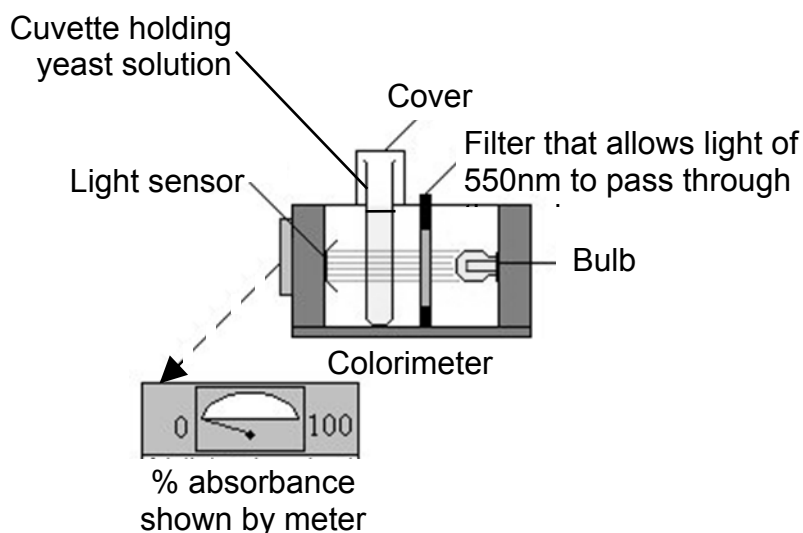


Fig. 4.1

Using this information and your own knowledge, design an experiment to test the hypothesis that:

“The rate of respiration in yeast cells is dependent on the concentration of sucrose solution.”

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- identify **independent** and **dependent** variables,
- describe the method with **scientific reasoning** used to decide the method so that the results are as accurate and reliable as possible,
- include layout of results **tables** and **graphs** with clear **headings** and labels,
- use the correct technical and scientific terms,
- include reference to safety measures to minimize any risks associated with the proposed experiment.

Your planning must be based on the assumption that you have been provided with the following equipment and materials which you must use:

- active yeast suspension
- 1.0 mol dm⁻³ glucose solution
- methylene blue
- test tubes
- stopwatch
- colorimeter
- 2 cuvettes
- beakers
- a variety of different sized measuring cylinders, syringes and pipettes for measuring volumes
-

Theoretical considerations (max 1m):

- During respiration, NAD⁺ and FAD are reduced to form NADH and FADH₂ during glycolysis, link reaction and Krebs cycle;
- Methylene blue replace NAD⁺ and FAD, it changes colour from blue to colourless when it is reduced;

Measurable Quantity (1m):

- The time taken for methylene blue to reach absorbance value of 15 a.u can be used to measure the rate of respiration;

Predicted Trend (1m):

- As concentration of glucose increases, time taken for methylene blue to decolorise and reach absorbance value of 15 a.u. decreases;
- Increase in frequency of effective enzyme-substrate collision between glucose and enzyme, increasing the rate of reaction;

Dependent variable (1m):

- Concentration of glucose solution/ moldm⁻³.

Independent variable (1m):

- Time taken for methylene blue to decolorise and reach 15 a.u of absorbance value/s.

Control (1m):

- Replace glucose solution with same volume of distilled water, subject to same experimental conditions to see that **decolorisation** is due to respiration when glucose is available;

Procedure:

- Carry out simple/ serial dilution of 1.0 mol dm⁻³ glucose solution with at least **5** different glucose concentration;
- Shows dilution table with correct calculation + correct headings + total volume same in all tubes + vol recorded to 1 dp;

Concentration of glucose/ mol dm ⁻³	Vol. of glucose solution/ cm ³	Vol. of distilled water/ cm ³
1.0	10.0	0.0
0.8	8.0	2.0

0.6	6.0	4.0
0.4	4.0	6.0
0.2	2.0	8.0

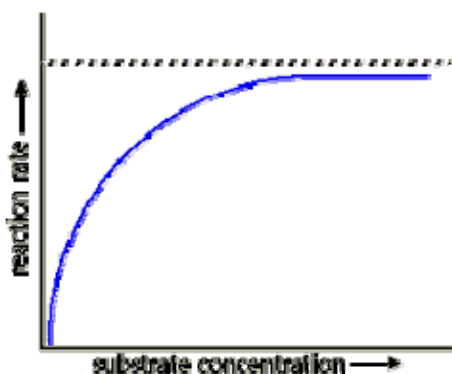
- Add stated volume (not exceeding 3 cm³) of methylene blue solution to stated volume (equal or more) of glucose solution;
- Ensure yeast suspension is stirred before adding to glucose solution to (scientific reasoning) ensure that yeast is well suspended/ not at the bottom of beaker;
- Add stated volume of yeast (equal volume/ more than glucose) to mixture;
Total volume should not exceed 15 cm³.
- Stir with a glass rod to (scientific reasoning) mix well;
- Pour mixture into cuvette provided and place cuvette into colorimeter (set to 550 nm);
- Start the stopwatch immediately and stop when absorbance reading on meter reaches 15 a.u.
- Record the time taken in a suitable table;
- Repeat experiment for the rest of the glucose solution;
- Rinse cuvette with distilled water and dry in between each measurement;
- Perform **2 replicates** for each glucose solution to ensure accuracy;
- **Repeat** the experiment **twice** to ensure reproducibility;

Results and Graph (2 m: 1m table; 1 m graph)

Concentration of glucose/ mol dm ⁻³	Time taken for methylene blue to decolorize/ s				Rate of respiration/ s ⁻¹
	Time 1	Time 2	Time 3	Average	
1.0					
0.8					
0.6					
0.4					
0.2					

- Table with appropriate headings and units;

Graph



- Graph with appropriate axes:
X-axis: concentration of glucose/ mold m-3
Y-axis: rate of respiration/ s-1
- Correct shape of graph sketched

Safety considerations:

- (**Precaution**) Dry your hands before switching the colorimeter on or off to (**risk**) avoid

	being electrocuted. • (<i>risk</i>) Methylene blue solution can be harmful / a skin irritant. (<i>Precaution</i>) Wear glove and goggles;	